

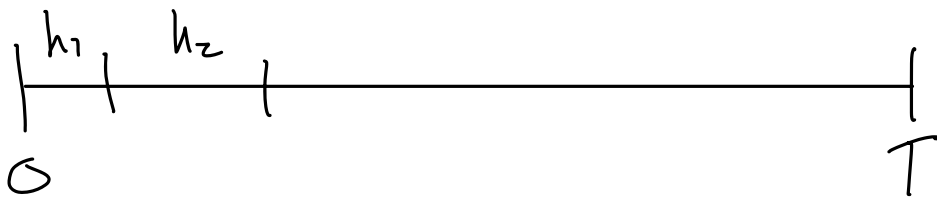
Error estimation and step size control

Given an IVP

$$u'(t) = f(u)$$

$$u(0) = u_0$$

$$0 \leq t \leq T$$



Two objectives:

1. Solve as accurately as possible
2. with as little computation as possible

1. suggests to use small h
2. suggests to use large h

We'd like to control global error.

There are methods to estimate global error a posteriori, but are expensive.

Usually we instead use local error control.

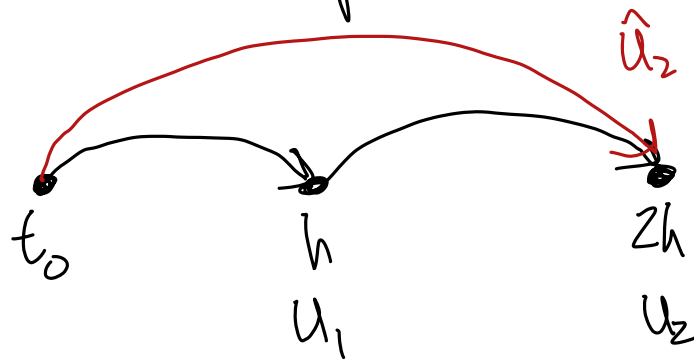
Choose a tolerance ε , and estimate the one step error at each step.

Goal: choose h_n so that $\delta_n < \varepsilon$.

Of course we always need to enforce absolute stability.

Estimating local error

Richardson extrapolation



$$e_1 = u(h) - u_1 = Ch^{p+1} + O(h^{p+2})$$

$$\begin{aligned} e_2 = u(zh) - u_2 &= (\text{new local error}) + (\text{propagated error}) \\ &= C(1 + O(h))h^{p+1} + e_1(1 + hf'(u) + O(h^2)) + O(h^{p+2}) \\ &= Zh^{p+1} + O(h^{p+2}) \end{aligned}$$

$$\hat{e}_2 = u(zh) - \hat{u}_2 = C(zh)^{p+1} = Cz^{p+1}h^{p+1}$$

So

$$u_2 - \hat{u}_2 = u(zh) + e_2 - (u(zh) + \hat{e}_2)$$

$$= e_2 - \hat{e}_2 = Ch^{p+1}(z - z^{p+1}) = \delta$$

Notice that if we take

$$u_2 + \frac{u_2 - \hat{u}_2}{2^p - 1}$$

Then the error will be

$$e_2 + \frac{e_2 - \hat{e}_2}{2^p - 1} = Ch^{p+1} \left(2 + \frac{2 - 2^{p+1}}{2^p - 1} \right) + O(h^{p+2})$$
$$= O(h^{p+2})$$

Step size selection

We want $\delta \approx \varepsilon$

We expect $\delta \approx Ch^{p+1}$

We want to choose h_* such that

$$Ch_*^{p+1} = \varepsilon$$

$$\left(\frac{h_*}{h} \right)^{p+1} = \frac{\varepsilon}{\delta} \Rightarrow h_*^{p+1} = \frac{\varepsilon}{\delta} h^{p+1}$$

$$h_* = \left(\frac{\varepsilon}{\delta} h^{p+1} \right)^{1/p+1} = \left(\frac{\varepsilon}{\delta} \right)^{1/p+1} h$$